Appendix D
Similar Model Systems

This appendix describes the models used throughout the text to describe mechanical, fluid flow, electrical, and acoustic systems. Figure D.1 shows the analog in the driving forces, currents, resistances, capacitance, and inductance in each of these models.

There are many examples of these models in the text. Chapter 4 covers the spring model of the elastic properties of materials (Fig. 4.3), the dashpot model of the viscous properties of materials (Fig. 4.48), the viscoelastic model of mechanical properties of materials, including the Maxwell (Fig. 4.52), Voigt (Fig. 4.57), and Kelvin/standard linear (Fig. 4.68) models. In Chap. 5, a mechanical model of muscles, with springs and dashpots (Fig. 5.6) is presented. Models of fluid flow are described in Chaps. 7 and 8. In fact, the Windkessel models of circulation in Chap. 8 are explicitly expressed in terms of electrical components (Fig. 8.56). The mechanical and flow model of breathing in Fig. 9.16b includes compliance, resistance, and inertance. Acoustic impedance (Fig. 10.19) and admittance (Fig. 10.20) are described in Chap. 10 (and in the problems in that chapter) in relation to mechanical analogs. In Chap. 10 there also are mechanical models of vibrations in the vocal tract (as vibrations in pipes and voice filtering theory) and vocal folds (vibrations in strings, (10.41), and mechanical model with mass, springs, and dashpots (Fig. 10.13) and the two-tube models of vowel formation (Fig. 10.25). In that chapter there is also a mechanical model of the outer and inner ears (Fig. 10.28), and the vibrations of the eardrum (Fig. 10.56), tapered, uncoiled cochlea (Fig. 10.38), and hair cells (Fig. 10.57). Axon nerve conduction in Chap. 12 involved a distributed model (See Sect. D.1).

Equivalent mechanical, electrical, and acoustic models are shown in Fig. D.2 of a typical system. The electrical model is described by

$$V = L\frac{\mathrm{d}I}{\mathrm{d}t} + IR + \frac{q}{C}, \tag{D.1}$$

for a voltage V producing a current I; $I = \mathrm{d}q/\mathrm{d}t$ where q is the charge. (The subscripts specific for the electrical model are omitted for simplicity.) The other models

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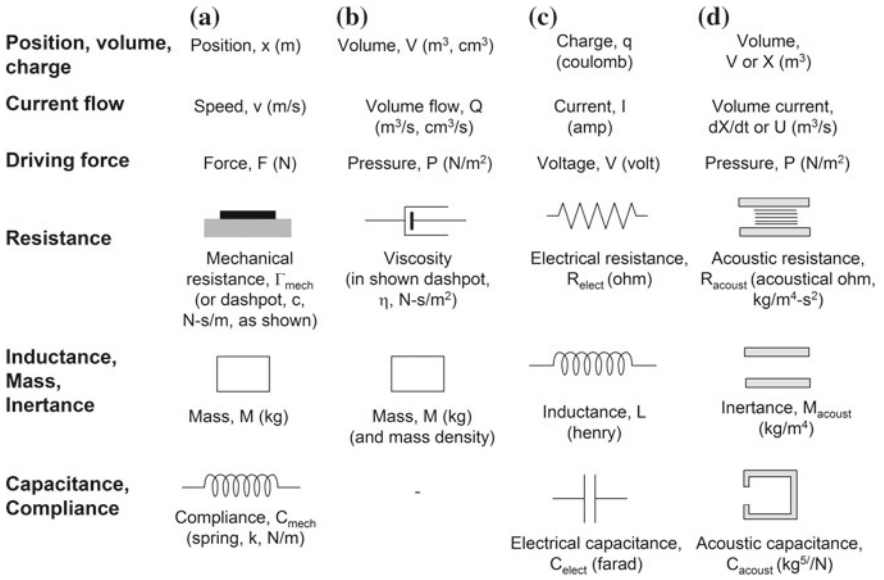


Fig. D.1 Model symbols are shown for **a** mechanical, **b** fluid flow, **c** electrical, and **d** acoustic models, along with the parameters and common units for each. The mechanical model is for linear (rectilinear) motion. Analogous parameters exists for the rotational mechanical model, such as for a pendulum. Viscosity is also important in mechanical models. Also see Table D.1 below (Based on [2, 3])

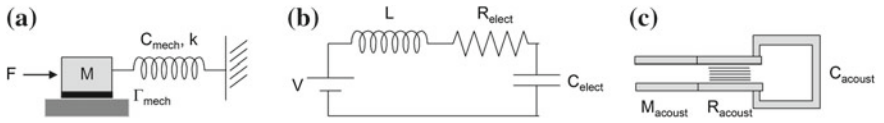


Fig. D.2 Equivalent **a** mechanical, **b** electrical, and **c** acoustic models (Based on [2, 3])

are described similarly, simply by changing the parameters. Equation D.1 also be written as the second-order differential equation

$$V = L\frac{\mathrm{d}^2 q}{\mathrm{d}t^2} + R\frac{\mathrm{d}q}{\mathrm{d}t} + \frac{q}{C}, \tag{D.2}$$

as in (C.30) and (C.34).

The general solution for an oscillating voltage $V(t) = V_0 \exp(i\omega t)$ is

$$I(t) = \frac{V_0 \exp(i\omega t)}{R + i\omega L + 1/(i\omega C)} = \frac{V(t)}{Z}, \tag{D.3}$$

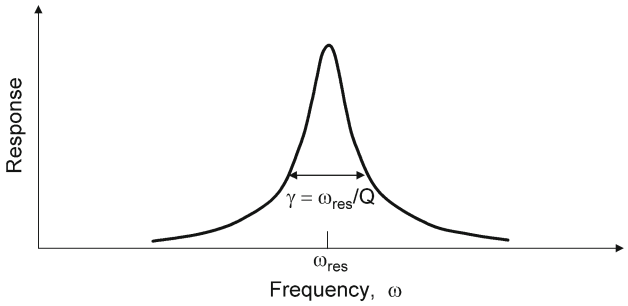


Fig. D.3 Resonant response

with complex impedance

$$Z = R + i\omega L + \frac{1}{i\omega C}. \tag{D.4}$$

The resonant frequency is seen in Fig. D.3 given by

$$\omega_{\text{res}} = \frac{1}{\sqrt{LC}} \tag{D.5}$$

in rad/s and $f_{\text{res}} = 1/(2\pi\sqrt{LC})$ in Hz or cps, and the quality factor Q is given

$$Q = \frac{\omega_{\text{res}} L}{R}. \tag{D.6}$$

Using the notation of (C.30) and (C.34)

$$Q = \frac{\omega_{\text{res}}}{\gamma}. \tag{D.7}$$

The full width of the resonance (between the points at half-maximum response) is ω_{res}/Q, as is illustrated in Fig. D.3. (This is actually the full width only for sharp resonances, for which $\omega_{\text{res}} \gg \gamma$ and $Q \gg 1$.) This full width is sometimes called the bandwidth (when expressed as f in Hz), Δf, and so an alternative definition of Q is $Q = f/\Delta f$; this is an equivalent definition in the low-loss, high Q limit.

After the excitation is turned off ((C.34) becoming (C.30)), the energy in the system exponentially decays to $1/e$ of the initial value in a time $t = 1/\gamma = Q/\omega_{\text{res}}$, where γ is the damping rate (as in (C.30)), which is R/L is here. This decay occurs in $Q/2\pi$ oscillation periods. This is consistent with the definition of Q as 2π(energy stored)/(energy dissipated per cycle).

D.1 Distributed Versus Lumped Models: Electrical Analogs of Blood Flow (Advanced Topic)

So far we have discussed lumped parameter models in this appendix. In (8.2) and (8.11) flow was analyzed with the vessel as a "lumped" parameter. We have also examined cases in this text in which the parameters are distributed per unit length, such as flow resistance per unit length for volumetric flow along an artery in Chap. 8 ((8.14) and (8.15)) and electrical resistance per unit length for current flow along an axon in Chap. 12 ((12.60) and (12.67)). These are "distributed" or "transmission-line" models.

A discretized version of the distributed electrical model is shown in Fig. 12.17. Let us say that each repeated section has (very short) length Δx. The changes in electrical voltage (the driving force) and current (the response) (Table D.1) along this length of an electrical cable are described by [566, 861]

$$V_{\text{elect}}(x) - V_{\text{elect}}(x + \Delta x) = -\frac{\partial V_{\text{elect}}}{\partial x}\,\Delta x = L_{\text{elect}}\frac{\partial I}{\partial t} + I R_{\text{elect}} \tag{D.8}$$

$$I(x) - I(x + \Delta x) = -\frac{\partial I}{\partial x}\,\Delta x = C_{\text{elect}}\frac{\partial V_{\text{elect}}}{\partial t} + \frac{V_{\text{elect}}}{R_{\text{elect}}}. \tag{D.9}$$

V_{elect} and I are functions of x and t. The resistance, inductance, and capacitance are those for this length Δx, and can also vary with x. These equations can be obtained using Kirchhoff's Laws (the 2nd and 1st laws, respectively). They were derived and then combined in the discussion of electrical signals along nerves in Chap. 12 (Fig. 12.17) to give the telegraph equations. Part of the first equation is Ohm's Law, with the voltage drop $\Delta V_{\text{elect}} = -(\partial V_{\text{elect}}/\partial x)\Delta x = I R_{\text{elect}}$.

The analogous equations for blood flow along a vessel of length Δx are:

$$P(x) - P(x + \Delta x) = -\frac{\partial P}{\partial x}\,\Delta x = L_{\text{flow}}\frac{\partial Q}{\partial t} + Q R_{\text{flow}} \tag{D.10}$$

$$Q(x) - Q(x + \Delta x) = -\frac{\partial Q}{\partial x}\,\Delta x = C_{\text{flow}}\frac{\partial P}{\partial t} + \frac{P}{R_{\text{flow}}} \tag{D.11}$$

Table D.1 Analog of blood flow and electrical circuits (with units)

Blood circulation parameter	Electrical parameter
Volume, V_{flow} (m^3)	Charge, q (C, coulomb)
Blood flow rate, Q (m^3/s)	Current, I (A, ampere)
Pressure, ΔP (N/m^2)	Voltage, V_{elect} (V, volt)
Vascular resistance, R_{flow} ($N\text{-}s/m^5$)	Resistance, R_{elect} (Ω, ohm)
Inertance, L_{flow} (kg/m^4)	Inductance, L_{elect} (H, henry)
Compliance, C_{flow} ($m^5/N\text{-}s$)	Capacitance, C_{elect} (F, farad)

where now the pressure is the driving force for the blood flow rate Q and R_{flow} is the vascular resistance. Without the inertance term, the first equation is just Poiseuille's Law (7.25), with the pressure drop $\Delta P = -(\partial P/\partial x)\Delta x = QR_{\text{elect}}$. The flow parameters are

$$R_{\text{flow}} = \frac{8\pi\eta L}{A^2} \tag{D.12}$$

$$C_{\text{flow}} = \frac{3LA(1 + r/w)^2}{Y(1 + 2r/w)} \tag{D.13}$$

$$L_{\text{flow}} = \frac{\rho L}{A}, \tag{D.14}$$

where A is the cross-sectional area of the vessel, L is its length (which is Δx for the discretized model), r is its radius, w is its wall thickness, and ρ is the blood mass density. These equations are useful for tracking blood flow within vessels with both resistive and compliant properties.

All models of materials (and systems and processes) can be improved mathematically by adding more terms, such as in the mechanical model in Fig. 5.13. The bigger issues are whether the elements in such simple or more complex models correspond to the physical components of the material. Even if they do not, it is still important to learn if the model can be used to predict operation correctly when conditions are changed.

References

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